

Define Problem Statements

1. Brainstorming & Ideation | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
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Problem Context

Banks and financial institutions receive thousands of credit card applications every day. Manual review becomes slow and error-prone when analysts must inspect income type, employment duration, credit history, overdue behavior, and demographic signals for every applicant. This project automates the initial approval screening using supervised machine learning and presents the prediction through a Flask web interface.

Traditional approval workflows depend heavily on manual judgment. At scale, this can delay decisions, create inconsistent review standards, and increase the chance of approving financially risky applicants.

Main Problem Statement

How might banks automate the initial credit card application screening process so that analysts can receive fast, consistent, and data-driven approval or rejection predictions while still keeping final human review available for borderline or high-risk cases?

Supporting Problem Statements

- How can applicant demographic and financial data be converted into clean machine learning features?
- How can multi-class credit payment status be simplified into clear risk labels for model training?
- How can a trained classification model be integrated into a user-friendly Flask application?
- How can the solution be deployed for real-time cloud prediction through IBM Watson Machine Learning?

Success Criteria

Area	Success Measure
Accuracy	The selected model performs better than simple baseline approval rules on validation data.
Speed	A single prediction is returned immediately after submitting the application form.
Usability	Bank staff or customers can enter required details without technical knowledge.
Scalability	The trained model can be saved, reused, and deployed to a cloud scoring endpoint.